

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment details.

1. System Requirements

System Name: ResuFit: Precision CV Matching System

Architecture Style: Microservices

Functional Requirements

- Submit CV
- View and Edit CV
- Answer Screening Questions
- Get Rating for CV
- Parse CV
- Compare CV with Job Requirements
- Determine Similarity
- Validate Applicant Responses
- Score Applicant Responses
- Generate Percentage-Based Result

Non-Functional Requirements

- Performance

Description: The system must respond to user input within a reasonable time frame.

- Reliability

Description: The system must be able to recover from failures and maintain data integrity.

- Security

Description: The system must protect user data and prevent unauthorized access.

- Privacy

Description: The system must ensure that user data is not shared without consent.

- Scalability

Description: The system must be able to handle a large number of users and requests.

- Usability

Description: The system must be easy to use and navigate.

- Maintainability

Description: The system must be easy to modify and update.

- Compatibility

Description: The system must be compatible with different browsers and devices.

- Integration

Description: The system must be able to integrate with other systems and services.

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, con-

```
System ResuFit:PrecisionCVMatchingSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices";
  Property source : String = "AI";
  Component CVParser = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Parse CV, Extract Key Information, Compare CV with Job Requi
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component ScreeningQuestionsModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Present Screening Questions, Validate Applicant Responses, S
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component RatingandScoringModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Determine Similarity, Generate Percentage-Based Result";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component UserInterfaceModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Submit CV, View and Edit CV, Answer Screening Questions, Get
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component NaturalLanguageProcessing(NLP)Module = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Parse CV, Determine Similarity";
```

```

    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component SimilarityCalculationModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Calculate Cosine Similarity, Calculate Jaccard Similarity";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component ChatGPTIntegrationModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Score Applicant Responses";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component RecruiterDashboardModule = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Define Job Requirements, View Applicant Responses, View Rati
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Attachment CVParser.out to ScreeningQuestionsModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment CVParser.out to RatingandScoringModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment ScreeningQuestionsModule.out to RatingandScoringModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment RatingandScoringModule.out to UserInterfaceModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment UserInterfaceModule.out to CVParser.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment UserInterfaceModule.out to ScreeningQuestionsModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment NaturalLanguageProcessing(NLP)Module.out to RatingandScoringModule.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}

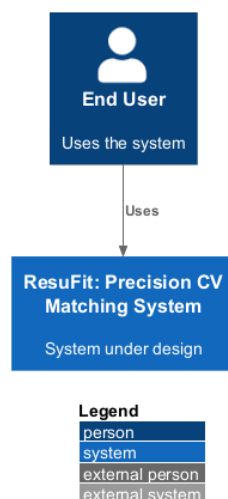
```

```

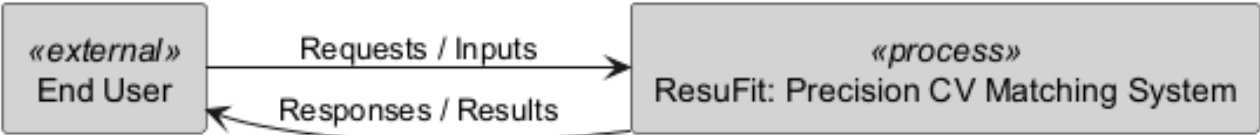
}
Attachment SimilarityCalculationModule.out to RatingandScoringModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment ChatGPTIntegrationModule.out to RatingandScoringModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Attachment UserInterfaceModule.out to ChatGPTIntegrationModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment UserInterfaceModule.out to RecruiterDashboardModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment ScreeningQuestionsModule.out to RecruiterDashboardModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = false;
  Property backpressure : String = "bounded";
}
Attachment RatingandScoringModule.out to RecruiterDashboardModule.in {
  Property delivery : String = "at-least-once";
  Property ordering : Boolean = true;
  Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

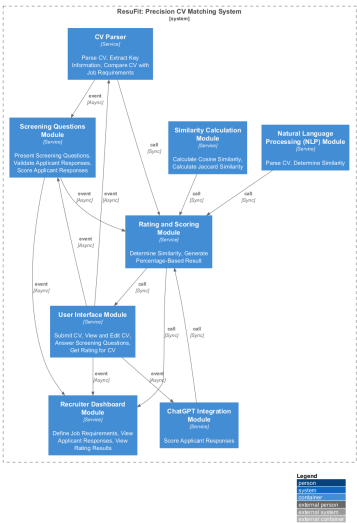
3. Context View



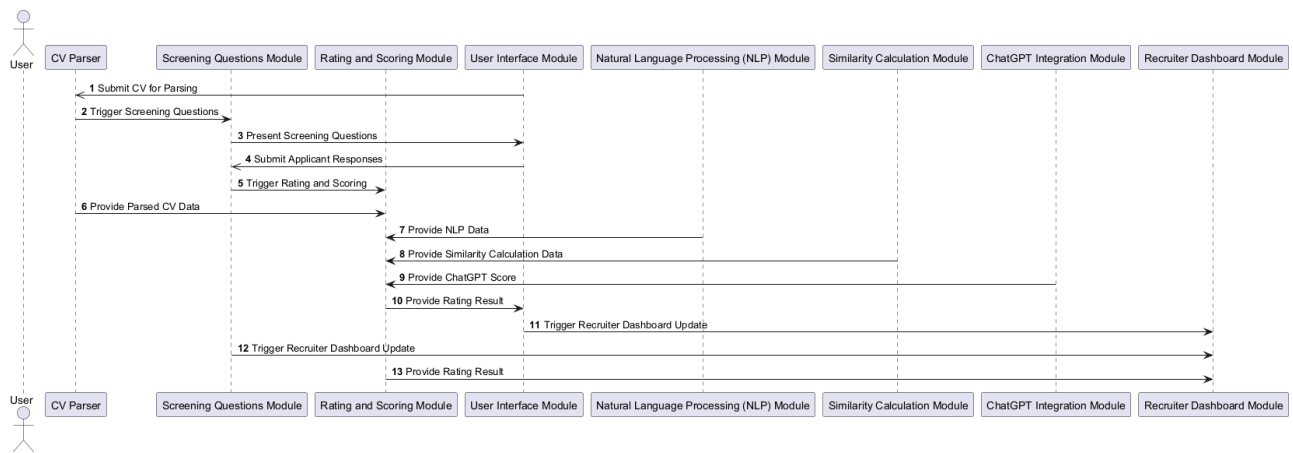
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

